성균관대학교
SUNGKYUNKWAN UNIVERSITY

PH.D. DISSERTATION DEFENSE

An Automated 3D Mapping and Point Cloud Semantic Modeling Framework for Spatial-Operation Robots

Sangmin Kim

Department of Electrical and Computer Engineering

Sungkyunkwan University

[Laboratory name] | [Defense date]

00 Outline

01 Introduction

02 Related Work

03 Proposed Framework

04 Experiments and Results

05 Conclusion and Future Work

01 PART

Introduction

Motivation, problem, and research objectives

IN THIS SECTION

01 Motivation

02 Problem definition

03 Research objectives

01 Research Background

- [Application context]
- [Current limitation]
- [Why this matters]

CONTEXT FIGURE

[Replace with one clear image]

KEY POINT [One sentence explaining the research need]

01 Problem Definition and Objectives

Problem

- [Limitation 1]
- [Limitation 2]
- [Limitation 3]

Objectives

- [Objective 1]
- [Objective 2]
- [Objective 3]

KEY POINT [Define the scope of this dissertation]

02 PART

Related Work

Position the research and identify the gap

IN THIS SECTION

01 Existing approaches

02 Comparative analysis

03 Research gap

02 Comparison of Related Work

Approach	Strength	Remaining gap
[Method A]	[Strength A]	[Gap A]
[Method B]	[Strength B]	[Gap B]
[Method C]	[Strength C]	[Gap C]

KEY POINT [The unresolved gap addressed by this work]

02 Research Gap

01 [Gap in acquisition]

02 [Gap in semantic modeling]

03 [Gap in downstream use]

03 PART

Proposed Framework

Present the core idea and method

IN THIS SECTION

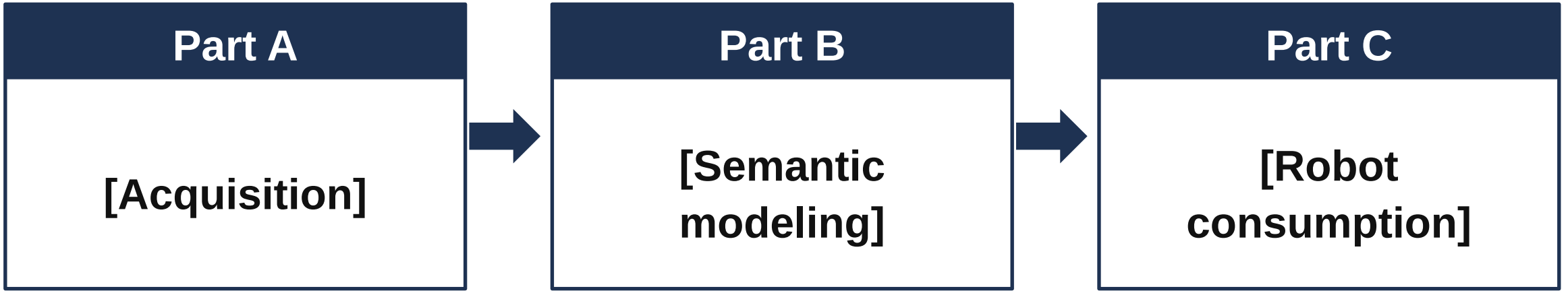
01 Automated mapping

02 Semantic modeling

03 Digital-twin integration

03 Framework Overview

[One integrated idea connecting the three parts]



KEY POINT	[State the principal contribution]

03 Part A — Automated Mapping

METHOD DIAGRAM

[Insert acquisition workflow]

Key mechanism

- [Input]
- [Decision rule]
- [Output]

KEY POINT [What is new in this component?]

03 Part B — Semantic Modeling

METHOD DIAGRAM

[Insert semantic workflow]

Key mechanism

- [Geometry]
- [Verification]
- [Graph update]

KEY POINT [What is new in this component?]

03 Part C — Digital-Twin Integration

SYSTEM DIAGRAM

[Insert twin and robot interfaces]

Key mechanism

- [Shared model]
- [Frame alignment]
- [Mission interface]

KEY POINT [State the demonstrated integration scope]

03 [Method Detail Title]

LARGE FIGURE OR EQUATION

[Use this layout for one key mechanism]

KEY POINT [One sentence interpreting the figure]

04 PART

Experiments and Results

Show evidence, controls, and limitations

IN THIS SECTION

01 Experimental setup

02 Quantitative evaluation

03 Discussion and limitations

04 Experimental Setup

- [Platform and sensor]
- [Environment or dataset]
- [Metric and comparison]

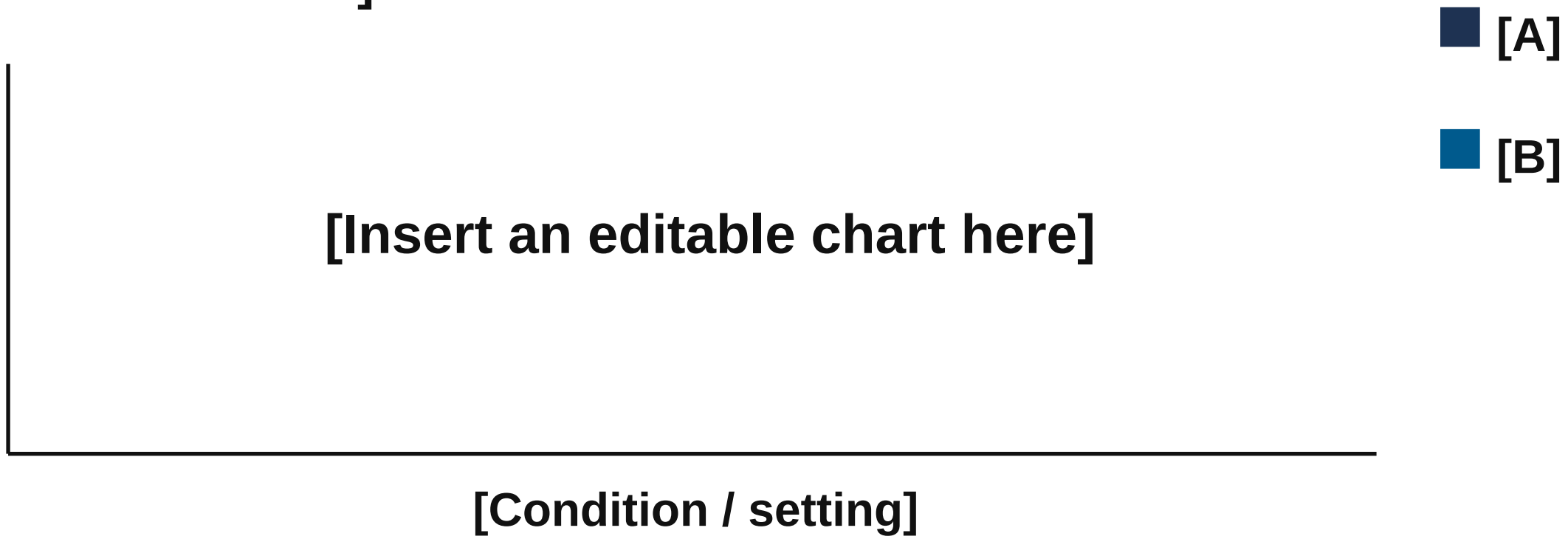
SETUP FIGURE

[Replace with experiment image]

KEY POINT [State the test scope and sample size]

04 Quantitative Results

[Metric and unit]



KEY POINT [Report the main finding with its uncertainty]

04 Qualitative Results

[Baseline]

RESULT FIGURE

[Proposed]

RESULT FIGURE

KEY POINT [Describe one visible difference under matched conditions]

04 Ablation Study

Configuration	Metric 1	Metric 2
[Baseline]	—	—
[+ Component A]	—	—
[+ Component B]	—	—

KEY POINT [Identify the contribution isolated by this experiment]

04 Discussion and Limitations

Supported findings

- [Finding 1]
- [Finding 2]
- [Finding 3]

Remaining limits

- [Limit 1]
- [Limit 2]
- [Limit 3]

KEY POINT [State how far the evidence supports the claim]

05 PART

Conclusion and Future Work

Close with contributions and next steps

IN THIS SECTION

01 Main contributions

02 Evidence and scope

03 Future work

05 Summary of Contributions

01 [Contribution to automated mapping]

02 [Contribution to semantic modeling]

03 [Contribution to robot integration]

05 Future Work

- [Method extension]
- [Independent validation]
- [Deployment study]

Questions and Discussion

Sangmin Kim

Sungkyunkwan University